

C1M5_UngradedLab_1

October 2, 2026

1 Module 5 Ungraded Lab 1: Multiparameter Models

Ungraded labs are designed to give learners an opportunity to apply theoretical material presented in previous videos. The goal of this ungraded lab is to conduct Bayesian inference on a multiparameter model. Ungraded labs can be completed when assigned or in conjunction with upcoming videos. Answers to ungraded labs will be provided in upcoming videos.

1.1 Problem #1 normal-inverse χ^2

The Hubble data frame contains 3 columns and 24 rows. The columns are:

- **Galaxy:** A (factor) label identifying the galaxy.
- **y:** The galaxy's relative velocity in kilometres per second.
- **x:** The galaxy's distance in Mega parsecs. 1 parsec is 3.09e13 km.

For this problem, let's just single out **x**. Suppose that **x** is a vector of realizations from $X_1, \dots, X_{24} \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Our goal will be to simulate from a joint posterior distribution over (μ, σ^2) given the data. Here's the data:

```
[1]: library(LaplacesDemon)
hubble = read.table(url("https://raw.githubusercontent.com/bzaharatos/STAT-5630/
↪main/hubble.txt"), sep = "\t")

n = length(hubble$x);

#frequentist point estimates
var(hubble$x)
mean(hubble$x)
#se of mean
var(hubble$x)/n
```

33.8101389492754

12.0545833333333

1.40875578955314

In class, we showed that the joint posterior distribution for (μ, σ^2) is normal-scaled-inverse- χ^2 . We also wrote down the conditional distributions for $\mu | \sigma^2, \mathbf{x}$ and for $\sigma^2 | \mathbf{x}$. We can use these conditional distributions to sample from the joint posterior distribution:

1 Draw σ^2 from $\sigma^2 \mid \mathbf{x}$

2 Draw μ from $\mu \mid \sigma^2, \mathbf{x}$, where σ^2 is from step 1.

Draw $m = 10,000$ samples from the joint posterior. Provide a histogram of the marginal distributions $\mu \mid \sigma^2, \mathbf{x}$ and $\sigma^2 \mid \mathbf{x}$.

[]:

Plot the histograms.

[]: